

Computer Networks

Lab Task 1: Networking Commands

The networking commands mentioned in the document were executed in Ubuntu Linux using Windows Subsystem for Linux (WSL). The commands helped in understanding network configuration, connectivity, routing, and packet capturing.

`sudo apt update`

- updates the list of available software packages
- the system updates the package repo information

`sudo apt install net-tools dnsutils whois tcpdump ethtool`

- installs networking utilities required for executing network commands

1. `ip addr`

- Displays IP address and interface information
- Shows network interfaces along with IP and MAC addresses

```
bhaavya@DESKTOP-H5EGEGB:/mnt/c/WINDOWS/system32$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:60:e8:f7 brd ff:ff:ff:ff:ff:ff
    inet 172.28.17.216/20 brd 172.28.31.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe60:e8f7/64 scope link
        valid_lft forever preferred_lft forever
```

2. `ip r`

- Displays the rerouting table
- Shows default gateway and routing paths

```
bhaavya@DESKTOP-H5EGEGB:/mnt/c/WINDOWS/system32$ ip r
default via 172.28.16.1 dev eth0 proto kernel
172.28.16.0/20 dev eth0 proto kernel scope link src 172.28.17.216
```

3. `host google.com`

- Resolves domain name to IP address
- The domain name is resolved to its IP address

```
bhaavya@DESKTOP-H5EGEGB:/mnt/c/WINDOWS/system32$ host google.com
google.com has address 142.251.220.46
google.com has IPv6 address 2404:6800:4009:80b::200e
google.com mail is handled by 10 smtp.google.com.
```

4. whois google.com

- Displays domain registration information
- Displays registrar, ownership, and registration details

```
bhaavya@DESKTOP-H5EGEB:/mnt/c/WINDOWS/system32$ whois google.com
Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-01-09T12:19:35Z <<<
```

5. cat /etc/services | more

- Displays standard network services and port numbers
- Lists well known services and their associated port numbers

```
bhaavya@DESKTOP-H5EGEB:/mnt/c/WINDOWS/system32$ cat /etc/services | more
# Network services, Internet style
#
# Updated from https://www.iana.org/assignments/service-names-port-numbers/s
service-names-port-numbers.xhtml .
#
# New ports will be added on request if they have been officially assigned
# by IANA and used in the real-world or are needed by a debian package.
# If you need a huge list of used numbers please install the nmap package.

tcpmux      1/tcp                # TCP port service multiplex
er
echo        7/tcp
echo        7/udp
discard    9/tcp                sink null
discard    9/udp                sink null
systat     11/tcp               users
daytime    13/tcp
daytime    13/udp
netstat    15/tcp
qotd       17/tcp                quote
chargen    19/tcp               ttytst source
chargen    19/udp               ttytst source
ftp-data   20/tcp
ftp        21/tcp
fsp        21/udp               fspd
ssh        22/tcp                # SSH Remote Login Protocol
telnet     23/tcp
smtp       25/tcp               mail
time       37/tcp               timserver
time       37/udp               timserver
whois      43/tcp               nicname
tacacs     49/tcp               # Login Host Protocol (TACAC
S)
tacacs     49/udp
domain     53/tcp               # Domain Name Server
domain     53/udp
bootps     67/udp
bootpc     68/udp
tftp       69/udp
gopher     70/tcp               # Internet Gopher
--More--
```

6. arp

- Displays the ARP table
- Shows IP and MAC address mapping of connected devices

```
bhaavya@DESKTOP-H5EGEG8:/mnt/c/WINDOWS/system32$ arp
Address          HWtype  HWaddress      Flags Mask    Iface
DESKTOP-H5EGEG8.mshome. ether    00:15:5d:01:1a:6b  C           eth0
```

7. netstat

- Displays active network connections
- Shows current network connections and their status

netstat -a | more

- Displays all listening and non-listening ports
- Shows all active TCP and UDP connections

```
bhaavya@DESKTOP-H5EGEG8:/mnt/c/WINDOWS/system32$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address          Foreign Address         State
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags   Type       State         I-Node  Path
unix  3      [ ]     STREAM    CONNECTED    11095    /run/dbus/system_bus_socket
unix  2      [ ]     STREAM    CONNECTED    19327
unix  3      [ ]     STREAM    CONNECTED    14720
unix  3      [ ]     STREAM    CONNECTED    14783    /run/dbus/system_bus_socket
unix  2      [ ]     DGRAM     CONNECTED    18502    /run/user/1000/systemd/notify
unix  3      [ ]     STREAM    CONNECTED    13757
unix  3      [ ]     STREAM    CONNECTED    15776    /run/systemd/journal/stdout
unix  2      [ ]     DGRAM     CONNECTED    15702    /run/user/0/systemd/notify
unix  3      [ ]     STREAM    CONNECTED    15575    /tmp/.X11-unix/X0
unix  3      [ ]     STREAM    CONNECTED    11855
unix  2      [ ]     DGRAM     CONNECTED    2257     /var/run/chrony/chronyd.sock
unix  2      [ ]     DGRAM     CONNECTED    12869
unix  3      [ ]     STREAM    CONNECTED    12752
unix  3      [ ]     STREAM    CONNECTED    15443
unix  2      [ ]     DGRAM     CONNECTED    12878
unix  3      [ ]     STREAM    CONNECTED    10198
unix  2      [ ]     STREAM    CONNECTED    13970
unix  3      [ ]     STREAM    CONNECTED    14955    /run/dbus/system_bus_socket
unix  3      [ ]     STREAM    CONNECTED    11097    /run/dbus/system_bus_socket
unix  3      [ ]     STREAM    CONNECTED    13939
unix  3      [ ]     DGRAM     CONNECTED    11728
unix  3      [ ]     DGRAM     CONNECTED    9140
unix  2      [ ]     DGRAM     CONNECTED    11836
unix  3      [ ]     DGRAM     CONNECTED    9139     /run/systemd/notify
unix  3      [ ]     STREAM    CONNECTED    10238
unix  3      [ ]     STREAM    CONNECTED    18507
unix  2      [ ]     DGRAM     CONNECTED    9148     /run/systemd/journal/syslog
unix  3      [ ]     STREAM    CONNECTED    12718
unix  3      [ ]     STREAM    CONNECTED    10094    /run/systemd/journal/stdout
unix  11     [ ]     DGRAM     CONNECTED    9156     /run/systemd/journal/dev-log
unix  8      [ ]     DGRAM     CONNECTED    9158     /run/systemd/journal/socket
unix  3      [ ]     STREAM    CONNECTED    18477
unix  2      [ ]     DGRAM     CONNECTED    11725
unix  3      [ ]     STREAM    CONNECTED    16495
unix  3      [ ]     STREAM    CONNECTED    10929
unix  3      [ ]     STREAM    CONNECTED    12233    /run/systemd/journal/stdout
unix  2      [ ]     DGRAM     CONNECTED    14782
unix  2      [ ]     DGRAM     CONNECTED    16521
unix  3      [ ]     STREAM    CONNECTED    11770    /run/systemd/journal/stdout
unix  2      [ ]     DGRAM     CONNECTED    14031
unix  3      [ ]     STREAM    CONNECTED    10093
unix  3      [ ]     STREAM    CONNECTED    12241
unix  3      [ ]     STREAM    CONNECTED    16490
unix  2      [ ]     DGRAM     CONNECTED    18487
unix  3      [ ]     STREAM    CONNECTED    17931
unix  3      [ ]     STREAM    CONNECTED    13898
unix  3      [ ]     DGRAM     CONNECTED    18504
unix  3      [ ]     STREAM    CONNECTED    10135    /run/systemd/journal/stdout
unix  3      [ ]     STREAM    CONNECTED    14848    /run/systemd/journal/stdout
unix  3      [ ]     STREAM    CONNECTED    13993    /run/dbus/system_bus_socket
unix  3      [ ]     STREAM    CONNECTED    10119    /run/systemd/journal/stdout
```

8. ss -tlp | more

- Displays listening TCP ports
- Shows TCP ports and the processes using them

```
bhaavya@DESKTOP-H5EGEB:/mnt/c/WINDOWS/system32$ ss -tlp | more
State  Recv-Q  Send-Q  Local Address:Port  Peer Address:Port Process
LISTEN 0        1000    10.255.255.254:domain  0.0.0.0:*
LISTEN 0        4096    127.0.0.53%lo:domain  0.0.0.0:*
bhaavya@DESKTOP-H5EGEB:/mnt/c/WINDOWS/system32$ |
```

9. sudo ethtool eth0

- Displays ethernet interface properties
- Displays speed, duplex mode, and link status

```
bhaavya@DESKTOP-H5EGEB:/mnt/c/WINDOWS/system32$ sudo ethtool eth0
[sudo] password for bhaavya:
Settings for eth0:
    Supported ports: [ ]
    Supported link modes:   Not reported
    Supported pause frame use: No
    Supports auto-negotiation: No
    Supported FEC modes:   Not reported
    Advertised link modes:  Not reported
    Advertised pause frame use: No
    Advertised auto-negotiation: No
    Advertised FEC modes:   Not reported
    Speed: 10000Mb/s
    Duplex: Full
    Auto-negotiation: off
    Port: Other
    PHYAD: 0
    Transceiver: internal
    Current message level: 0x000000f7 (247)
                          drv probe link ifdown ifup rx_err tx_err
    Link detected: yes
```

10. cat /proc/net/sockstat

- Displays socket usage stats
- Shows number of sockets used by TCP and UDP

```
bhaavya@DESKTOP-H5EGEB:/mnt/c/WINDOWS/system32$ cat /proc/net/sockstat
sockets: used 215
TCP: inuse 3 orphan 0 tw 0 alloc 3 mem 741
UDP: inuse 3 mem 0
UDPLITE: inuse 0
RAW: inuse 0
FRAG: inuse 0 memory 0
```

11. cat /proc/net/tcp

- Displays TCP socket information and current TCP connections

```
bhaavya@DESKTOP-H5EGEB:/mnt/c/WINDOWS/system32$ cat /proc/net/tcp
sl  local_address rem_address  st tx_queue rx_queue tr tm->when retrnsmt  uid  timeout inode
0:  FEFFFF0A:0035 00000000:0000 0A 00000000:00000000 00:00000000 00000000  0    0 6154 1 0000000000000000 100 0 0 10 0
1:  3500007F:0035 00000000:0000 0A 00000000:00000000 00:00000000 00000000  101   0 13941 1 0000000000000000 100 0 0 10 5
2:  D8111CAC:0018 81A0F322:01B5 06 00000000:00000000 03:00000585 00000000  0    0 0 3 0000000000000000
bhaavya@DESKTOP-H5EGEB:/mnt/c/WINDOWS/system32$ |
```

12. cat /proc/net/udp

- Displays UDP socket information and socket details

```
bhaavya@DESKTOP-H5EGEG8:/mnt/c/WINDOWS/system32$ cat /proc/net/udp
sl local_address rem_address  st tx_queue rx_queue tr tm->when retrnsmt  uid  timeout inode ref pointer drops
1196: 3500007f:0035 00000000:0000 07 00000000:00000000 00:00000000 00000000 101 0 13940 2 0000000000000000 0
1196: ffffffff:0035 00000000:0000 07 00000000:00000000 00:00000000 00000000 0 0 6153 2 0000000000000000 0
1466: 0100007f:0143 00000000:0000 07 00000000:00000000 00:00000000 00000000 0 0 2254 2 0000000000000000 0
bhaavya@DESKTOP-H5EGEG8:/mnt/c/WINDOWS/system32$ |
```

13. sudo tcpdump -i eth0 -c 5

- Captures live network packets

```
bhaavya@DESKTOP-H5EGEG8:/mnt/c/WINDOWS/system32$ sudo tcpdump -i eth0 -c 5
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
18:06:27.323523 IP6 fe80::ae0d:e4af:462d:b85d > ff02::1:ff5d:cb2f: ICMP6, neighbor solicitation, who has fe80::7b8:f3f7:6d5d:cb2f, length 32
18:06:28.322329 IP6 fe80::ae0d:e4af:462d:b85d > ff02::1:ff5d:cb2f: ICMP6, neighbor solicitation, who has fe80::7b8:f3f7:6d5d:cb2f, length 32
^C
2 packets captured
2 packets received by filter
0 packets dropped by kernel
bhaavya@DESKTOP-H5EGEG8:/mnt/c/WINDOWS/system32$ |
```

14. sudo tcpdump -i eth0 -c 5 -w ~/test.pcap

- Captures network packets and stored them in a file

```
bhaavya@DESKTOP-H5EGEG8:/mnt/c/WINDOWS/system32$ sudo tcpdump -i eth0 -c 5 -w ~/test.pcap
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
5 packets captured
6 packets received by filter
0 packets dropped by kernel
bhaavya@DESKTOP-H5EGEG8:/mnt/c/WINDOWS/system32$
```

15. uptime

- Displays system running time
- shows system uptime and load average

16. ip neigh

- Displays neighbor ARP table
- shows neighbor IP and MAC address entries

17. hostname -I

- Displays system IP address assigned to the system

```
bhaavya@DESKTOP-H5EGEG8:/mnt/c/WINDOWS/system32$ uptime
18:19:06 up 56 min, 2 users, load average: 0.03, 0.09, 0.05
bhaavya@DESKTOP-H5EGEG8:/mnt/c/WINDOWS/system32$ ip eigh
Object "eigh" is unknown, try "ip help".
bhaavya@DESKTOP-H5EGEG8:/mnt/c/WINDOWS/system32$ ip neigh
172.28.16.1 dev eth0 lladdr 00:15:5d:01:1a:6b STALE
bhaavya@DESKTOP-H5EGEG8:/mnt/c/WINDOWS/system32$ hostname -I
172.28.17.216
bhaavya@DESKTOP-H5EGEG8:/mnt/c/WINDOWS/system32$ |
```

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